Underscope: Channel-Routing Intelligence — Research Plan, Requirements and Architecture

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Executive summary

Build a routing engine, not a price predictor. After a VIPER scan, the system ranks the channels a vehicle could go to (ACV dealer wholesale, Copart salvage, Copart's fleet and bank lane, a recycler, a direct buyer) by expected net proceeds, probability the sale completes, time to cash and post-sale risk, then names the buyer segment most likely to pay. A good buyer is the one with the highest expected net to the seller after fees, transport and unwind risk, with a completion probability above a floor, not the highest bidder.

Three facts shape everything below:

- **ACV and Copart are now one company.** Copart completed its $1.9 billion, $10.50-per-share acquisition of ACV on 16 Oct 2026; ACV runs as an independent subsidiary. Copart describes the combination as a remarketing platform spanning trade-ins, wholesale and salvage. The channel question this doc answers is now an internal problem for them, which is both the biggest opportunity and the biggest platform risk.
- **The volumes are large and already segmented.** Copart sold more than 4 million units in its last fiscal year, 81% of them from insurers. ACV sold 829,276 units in 2025 through 14,905 sellers and 22,062 buyers. Copart's fleet, bank and rental lane (Blue Car) grew about 20% in its latest quarter, which is where mis-routed vehicles most likely sit today.
- **We cannot get their data before a prototype exists, and we cannot price repairs precisely.** So the mechanism is built on proxy data (public Copart results, VIN decode, price indexes, our own scans), synthetic outcomes and a backtest harness, with a published data contract so their data plugs in later. The output is a ranked recommendation with a confidence band and a short list of what to verify, never a single number.

The doc contains seven research-agent briefs (section 3) with the findings each produced (sections 4 to 7), the buyer scorecard (8), the requirements (9), the architecture (10), the no-data build plan (11), costs (12), readiness and risks (13) and a 90-day plan with the

questions to put to Copart/ACV (14).

The question, restated

The decision is "where does this vehicle earn the most for its seller, net and risk-adjusted, and who is the buyer there", taken once per vehicle right after the scan. VIPER and its predecessors already produce the condition facts; what nobody produces is the comparison across channels that now sit inside one company.

Why "predict the price" is the wrong framing:

- A price point will be wrong often, and the first wrong number costs you the seller's trust. A ranking only has to get the order right, and it can say when two channels are too close to call.
- Repair cost cannot be estimated precisely from photos (frame, mechanical and flood damage are invisible), so any channel that depends on a repair estimate must carry a wide band. The ranking can still be confident when the gap between channels is larger than the band.
- Routing is a treatment-effect problem. For each past vehicle you only observe the channel it went to, never the counterfactual. The mechanism therefore has to be built around explicit assumptions, simulation and controlled experiments, not a single supervised model.

What the engine must answer for every scanned vehicle:

1. Which channels are legally and contractually available (title type, state rules, seller agreements)?
2. For each available channel, what is the distribution of net proceeds to the seller, the probability the sale completes without unwind, and the days to cash?
3. Which buyer segment inside the winning channel is the likely payer (dealer by region and inventory gap, rebuilder, dismantler, exporter, fleet buyer), and how reliable is that segment?
4. How confident is the ranking, and which single piece of missing information would change it?
5. What should a human verify before the vehicle is assigned?

Research agents

Eight briefs below. Each was run for this doc (findings in sections 4 to 7 and 12 to 13) and is written so it can be handed to a Claude Code sub-agent unchanged. Run agents 1 to 7 in parallel, each writing a `findings-<n>.md` with every claim linked to a page it opened; run agent 8 last against the other seven. Agents must mark any number they could not source as "unverified" rather than drop it or guess.

1. **Customer and volume agent.** Goal: name every seller type that makes a channel decision for damaged or off-cycle vehicles, with annual volumes. Questions: how many units do insurers, fleets, rental companies, banks, franchise and independent dealers dispose of each year, through which channels, and who inside each decides? Where do decisions get made by rule today (for example, every total loss goes to salvage)? Sources: Copart and ACV 10-Ks and earnings calls, Manheim and Cox Automotive reports, NADA data, fleet remarketing trade press (Auto Remarketing, Automotive Fleet). Deliverable: a segment table (segment, units per year, current channel, who decides, what they optimize). Done when every segment has a sourced volume or an explicit "unknown".
2. **Analog and incumbent agent.** Goal: catalogue every product that already decides a channel or predicts a channel price. Questions: what do ACV MAX, Copart IntelliSeller and ProQuote, vAuto ProfitTime GPS, Cox Upside, OPENLANE tools, Manheim MMR and recycler bid tools (Car-Part Bidmate, reParta) take as input, output and charge? Which of them compare channels, and which assume one channel? Sources: product pages, pricing pages, demo videos, dealer forums, patents. Deliverable: a comparison table plus a one-paragraph statement of the gap. Done when each tool has inputs, outputs, price and "channels compared" filled or marked unknown.
3. **Outside-opinion agent.** Goal: capture what dealers, remarketers, analysts and buyers say about AI inspection, AI pricing and "intelligence platforms". Questions: do dealers trust AI condition reports; what do they say about arbitration; what did analysts say the Copart-ACV combination is for; where have algorithmic pricing businesses failed (Vroom, Shift, Carvana's early years) and why? Sources: earnings transcripts, analyst notes, Dealer Refresh and Reddit dealer threads, Auto Remarketing, Automotive News, S&P and Cox commentary. Deliverable: a list of recurring objections with quotes and the counter-evidence for each. Done when at least ten distinct objections are sourced.
4. **Data agent.** Goal: list every dataset usable before Copart/ACV share anything, with license, cost and what it covers. Questions: which public pages expose Copart sold prices and lot attributes; which APIs decode VINs (NHTSA vPIC), price vehicles (Black

Book, J.D. Power, KBB, Marketcheck) and price parts (Car-Part, eBay); what do the sites' terms allow; what does a sample of 1,000 Copart lots look like? Sources: API docs, terms of use, a scraped-by-hand sample. Deliverable: a data inventory table and the data contract (section 7) listing exactly which fields we need from Copart/ACV. Done when every field in the contract has a proxy or an explicit gap.

5. **Decision-logic agent.** Goal: specify the math. Questions: fee schedules and seller fees on ACV and Copart; typical days to sale; arbitration and unwind rates by channel; transport cost by distance; how buyer segments differ in what they pay for (exporters, dismantlers, rebuilders, dealers). Sources: fee pages, 10-K risk sections, ACV arbitration policy, Copart member terms, recycler trade press. Deliverable: the net-proceeds formula per channel with every parameter sourced or flagged. Done when the formula runs end to end on a sample vehicle with ranges.
6. **Cost and architecture agent.** Goal: price the operation. Questions: per-vehicle inference cost on API models versus a self-hosted model; data licensing cost; human review minutes per vehicle; infrastructure for 1,000 and 100,000 vehicles a month. Sources: model pricing pages, cloud GPU pricing, BLS wages, data vendor quotes. Deliverable: a cost table at two volumes. Done when every line has a source or a quote request logged.
7. **Readiness and go-to-market agent.** Goal: judge whether Copart/ACV would adopt this and how a startup gets in. Questions: what have they acquired and partnered on (Monk, MAX Digital, ClearCar, Purple Wave); what pilots have they run with startups; what did their deal announcement promise about cross-channel selling; what metrics would their teams need to see? Sources: press releases, deal filings, conference talks, LinkedIn posts by their product leaders. Deliverable: a pilot proposal and a list of named counterparts. Done when the pilot has a metric, a sample size and a duration.
8. **Red-team agent.** Goal: argue that this should not be built, using agents 1 to 7's findings. Questions: why would Copart build it internally within a year; what makes the routing gain too small to pay for; what legal or contractual constraint kills the use case; where could a wrong recommendation cause a loss we would be blamed for? Deliverable: the five strongest objections and what evidence would settle each. Done when each objection has a test, not a rebuttal.

Findings: who needs this (Agent 1)

The strongest customer is the combined Copart/ACV itself, followed by fleet, rental and finance sellers; insurers are contract-bound and dealers are already served by Cox and ACV tools. Today vehicles are routed by who consigns them, not by what they are: Copart

runs separate lanes for rental returns, repo and off-lease units, dealer cars and insurance totals, and ACV runs dealer-to-dealer wholesale.

Segment	Volume (sourced)	Channel today	Who decides	What they optimize
Insurers (total losses)	81% of Copart's 4M+ units a year (Copart 10-K FY2025)	Salvage auction under contract	Claims and salvage teams, by total-loss rule	Salvage return, cycle time
Fleet, rental, bank and finance	Blue Car lane up about 20% in Q4 FY2026 (May to Jul 2026); fleet and finance volume growing at a double-digit pace while rental sellers repair more (Q4 call, Q3 call)	Copart Rental and Wholesale auctions, Manheim, OPENLANE, ACV commercial	Remarketing manager	Net return, days to sale, repair-or-sell
Franchise and independent dealers	ACV: 829,276 units, 14,905 sellers, 22,062 buyers in 2025 (ACV 10-K 2025); Copart dealer units up 5.8% in Q4 FY2026	ACV, Manheim, OPENLANE, Copart dealer lanes, Cox Upside	Used-car manager	Gross per unit, inventory turn, no arbitration

Segment	Volume (sourced)	Channel today	Who decides	What they optimize
Copart/ACV internally	Copart sold more than 4M units in its last fiscal year; 25 dedicated wholesale facilities serve 80% of the addressable wholesale market (Q4 call)	Lane assignment by seller type and contract	Yard and lane rules, seller agreements	Revenue per unit, buyer liquidity, arbitration cost

What the volumes say about demand:

- Copart's CEO said the ACV combination could broaden access to franchise dealers and higher-end trade-ins while supporting demand for lightly damaged insurance vehicles and international sales. That is the routing problem stated by the buyer: some insurance cars belong in the dealer channel, and some trade-ins belong with Copart's international buyers.
- Buyers already cross channels. Over the past three years a majority of the 30,000-plus members who entered Copart through non-insurance vehicles bid on an insurance vehicle within 90 days, so a buyer-fit model has a real population to learn from.
- Rental sellers are choosing repair over sale more often (Q3 FY2026 call), which means the fix-versus-sell decision is live and currently made with the seller's own rules.
- Total-loss frequency hit 23.3% in Q2 FY2026 (Nov 2025 to Jan 2026), a record for that quarter, so the supply of damaged-but-sellable vehicles keeps rising while U.S. unit volume fell 5.7% in Q4 FY2026. Routing that improves revenue per unit is worth more to Copart when units are flat.

Seller types Copart names on its own site: insurance companies, financial institutions, dealers, rental car companies, charities, fleet operators and individuals; buyer types: dealers, dismantlers, rebuilders, exporters and the public (Copart release, 16 Oct 2026).

Findings: what already exists (Agent 2)

Every tool on the market prices a vehicle for one channel or compares at most two, and each assumes its owner's marketplace; none ranks wholesale, salvage and parts side by side with a buyer-segment view. That is the gap, and the deal announcement shows Copart intends to fill it from the inside.

Tool	Owner	Input	Output	Channels compared	Price
ACV MAX Recommendations and VIPER	ACV (Copart)	VIN, VIPER scan (360 exterior, undercarriage, damage detection, tire tread), MAX inventory data	Retail pricing guidance, wholesale guarantee, consumer offer	Retail vs ACV wholesale	Dealer subscription; not public
ProQuote and IntelliSeller	Copart	Images, claim data	Salvage value estimate, repair-or-total support, minimum bids	Copart salvage only	Bundled for sellers
vAuto ProfitTime GPS	Cox	Appraisal, market data	Investment value tier, retail and acquisition price	Retail vs wholesale, by acquisition channel	Dealer subscription
Cox Upside	Cox	KBB Instant Cash Offer appraisal	Guaranteed minimum wholesale price plus profit share; 500+ dealers, about $30M shared back	Retail vs Cox wholesale	Fee per sale; not public
Car-Part Bidmate	Car-Part.com	Copart lot imports, recycler's sales history and stock	Value of each lot to that recycler, damaged parts discounted	Parts only (buyer side)	Recycler subscription
AutoEstimatePro, sendit scan, TopFuel	Startups	Listing photos, VIN	Repair cost, max bid, flip verdict	Buyer side, salvage only	Free to $99 a month

Three things follow:

- The condition layer is solved by the incumbents. VIPER captures 360-degree exterior imagery, undercarriage images, automatic damage detection and tire tread in under 60 seconds, built on more than one million inspections a year and 250-plus condition data points per vehicle. We should consume that output, not rebuild it.
- The guarantee is what sellers actually buy. Upside and ACV's guaranteed sale (18% of ACV units by Q3 2025, per its Q3 2025 call) convert because they remove the seller's market risk. A routing recommendation becomes a product when it can be backed by a floor price, which needs the channel price distributions this doc specifies.
- Nobody owns the comparison. Copart's CEO described the goal on the Q4 FY2026 call as connecting the right vehicle to the right buyer through the right channel without forcing every vehicle into a single operating model (Repairer Driven News), and said ACV cars will keep selling on ACV's site using Copart logistics. The brands stay separate; the brain that decides between them does not exist yet.

Findings: what outsiders think (Agent 3)

The market does not doubt that AI can read a car; it doubts that anyone will stand behind the result. Every recurring objection is about trust, arbitration and who eats the loss when the machine is wrong.

Objection	Where it shows up	What it means for the design
Condition reports and arbitration favor the auction, not the buyer	Dealer threads on Manheim and reviews of ACV and OPENLANE repeatedly cite green-light cars with undisclosed faults and small arbitration credits	Every recommendation ships with the photos and measurements behind it; a buyer-risk score is part of the buyer definition
A visual inspection has limits by definition	The NAAA position statement updated 1 Jun 2026 defines a condition report as a visual inspection with stated limitations (NAAA)	Mechanical, frame and flood risk are modeled as unknowns with a price band, never asserted

Objection	Where it shows up	What it means for the design
Guarantees cost more than they earn when inspection is wrong	ACV's assurance revenue grew 27% to $81.6M in 2025 while its assurance cost of revenue grew 30% to $73.3M (analysis)	Routing should send high-dispute-risk vehicles to as-is channels and keep guaranteed channels for low-risk cars; this is a direct, measurable win for Copart/ACV
Algorithms do not fix operations	Vroom shut its online used-car business in 2024 after reconditioning and acquisition costs overwhelmed its pricing (post-mortem)	Do not take inventory or promise a price; recommend, then measure realized net
The incumbents will build it themselves	Copart's deal rationale names a combined demand engine that gets the right vehicle to the right buyer (Copart release, 10 Sep 2026); Barrington called the deal a long-term growth catalyst, Needham moved ACV to hold (coverage)	Build the mechanism they would otherwise build, with the evaluation harness they lack, and sell it as a plug-in or a pilot rather than a competitor
Volume is the structural risk, not technology	An investor note on Copart flags fewer accidents from driver assistance and a rising total-loss share as the two long-term risks (Parqet summary)	Revenue-per-unit tools get more valuable as units flatten, which is the pitch to Copart

Two quotes worth keeping for the pitch: an auction executive told Auto Remarketing that a good photograph is often worth more than a condition report (ServNet remarks), and ACV's own buyer terms put buyers with a 25% arbitration rate or $250 average arbitration cost into a limited program (ACV terms), which confirms that buyer-level dispute behavior is already tracked and is the right raw material for a buyer score.

Findings: data we can and cannot get (Agent 4)

Enough public and licensed data exists to build and test the mechanism on the salvage side; the dealer-wholesale side (ACV prices, condition reports, arbitration outcomes) is only available from ACV, which is why the prototype must run on proxies and then accept their data through a fixed contract.

Data	Source now	Covers	License and cost	Gap
Copart lot attributes and final bids	Copart members can download CSV sales data; Copart's FAQ says it does not keep sold-price data, but third parties archive final bids (Copart FAQ, Bid.cars archive scraper, Apibara API)	Year, make, model, VIN, damage, title brand, odometer, run status, yard, sale date, final bid, estimated retail value	Scraped data; terms of use need a lawyer's read before commercial training	Buyer identity and buyer type are not exposed
VIN decode	NHTSA vPIC API	Build, trim, engine, plant	Free, public	No options or color
Wholesale and retail values	Black Book, J.D. Power, KBB, Marketcheck APIs; Manheim index for trend	Clean-title values by trim, mileage and region	Licensed; quotes needed	Branded-title adjustments are thin; measure them ourselves
Rebuilt and salvage discounts	J.D. Power guidance and a matched-listing study (J.D. Power, forcar.org)	20 to 40% typical; about 17.6% in matched listings	Public	Varies by model; needs our own measurement

Data	Source now	Covers	License and cost	Gap
Parts demand and prices	Car-Part.com, eBay sold listings, Hollander interchange (licensed to the industry, about 155 parts and assemblies per vehicle)	Part-level prices and sell-through	Licensed or scraped	Sell-through time is the weak field
Our own scans	Underscope uploads: exterior, interior, engine bay, underbody photos plus VIN	Condition labels from our review flow	Owned	Small at first; grows with every review
Title and state rules	State DMV statutes (for example Texas, Nevada)	Which channels a title brand allows	Public	Must be encoded state by state

What only Copart/ACV hold: ACV sale prices and bids by buyer, condition reports with inspector labels, arbitration claims and outcomes, buyer histories across both platforms, seller contracts and fee schedules, transport costs by lane, and the VIPER output itself. We do not need all of it on day one; we need a contract for it.

The data contract (what we ask for, per vehicle and per transaction):

1. Vehicle: VIN, odometer, title brand and state, seller type, intake location, VIPER condition summary (damage list with part, type, severity, image references), tire tread, undercarriage flags.
2. Listing and sale: channel, lane, auction date, reserve or guarantee, bid history (count, top three bids, bidder segment), final price, fees charged, days from intake to sale, transport miles.
3. Buyer: anonymized buyer id, segment (dealer, rebuilder, dismantler, exporter, public), home region, lifetime purchases, arbitration rate, payment-default rate, export share.
4. Post-sale: arbitration opened (yes/no), reason, resolution, credit paid, unwind (yes/no), relist outcome.

5. Counterfactual hooks: for a small random sample, the channel the rules would have chosen and the channel actually used, so routing lift can be measured without bias.

Each field above is mapped to a proxy in the prototype (section 11) or marked as a gap the pilot fills.

Definition of a good buyer

A good buyer for a given vehicle is the buyer segment whose expected net to the seller is highest after fees, transport, holding time and post-sale risk, with a completion probability above a floor the seller sets. The highest bid is not the test; a bid that is later arbitrated, unwound or unpaid is worth less than a lower bid that closes.

The definition works at two levels. At the channel level, an auction lane is a mix of buyer segments, so choosing the lane is choosing the mix; the engine picks the lane whose buyer mix maximizes expected net for this vehicle. At the buyer level, for direct offers or invitation-only sales, the same scorecard ranks named buyers.

```
\mathrm{Net}_{c} = \mathbb{E}[\mathrm{Price}_{c}] - \mathrm{Fees}_{c} -
\mathrm{Transport}_{c} - \mathrm{Hold}_{c}(d) - P_{c}
(\mathrm{dispute})\,\mathbb{E}[\mathrm{credit}] - P_{c}
(\mathrm{unwind})\,\mathbb{E}[\mathrm{relist\ loss}]
```

where c is a channel (or a buyer segment inside it), d is expected days to cash, and every term carries a range, not a point.

Dimension	What it measures	Proxy in the prototype	Signal once Copart/ACV data plugs in
Willingness to pay for this vehicle	Price distribution a segment pays for this make, model, age, damage and title	Copart final bids by lane and damage type; clean-value APIs; parts prices for dismantler value; export demand by model from international bid share	Bid histories by bidder segment on both platforms

Dimension	What it measures	Proxy in the prototype	Signal once Copart/ACV data plugs in
Probability of completion	Bid turns into payment and pickup	Assume lane-level averages; flag as unverified	Buyer-level close rate, payment-default rate
Post-sale risk	Chance of arbitration or unwind and its cost	Channel rules: Copart sells as-is, ACV offers a 10-day arbitration window and caps remedies at price paid (ACV policy)	Buyer arbitration rate and average credit; ACV already thresholds buyers at 25% rate or $250 average cost
Time to cash	Days from assignment to funds	Lane calendars (Copart rental and wholesale auctions run twice weekly), ACV 20-minute auctions, parts 6 to 18 months	Measured per lane and buyer segment
Cost to reach	Transport, fees, export paperwork	Distance to yard and buyer region; public buyer-fee tiers	Actual transport lanes and seller contracts
Fit	Does this segment want this vehicle more than others do	Dismantlers for high-demand parts models, rebuilders for light damage on clean-title cars, exporters for models with foreign demand, dealers for retail-ready units	Crossover behavior: Copart reports most non-insurance entrants bid on insurance cars within 90 days

Decision rules on top of the score:

- Rank channels by expected net; show the top two with their bands. If the bands overlap by more than a threshold (start at 15% of the lower band), declare "too close to call" and name the one check that would separate them.

- Apply the seller's preference as weights: a fleet remarketer may trade $200 of expected net for ten fewer days to cash; a dealer may refuse any channel with arbitration exposure above a limit.
- Exclude channels the title or state law forbids before scoring (non-repairable titles cannot be rebuilt or registered in Texas and must go to a licensed wrecker in Nevada).
- Route high-dispute-risk vehicles (hidden mechanical or frame unknowns) to as-is channels unless the price gain in a guaranteed channel exceeds the expected dispute cost.

Platform requirements

The platform takes a scanned vehicle and returns a ranked channel recommendation with a buyer segment, a net-proceeds band, a confidence level and a verification list, within 60 seconds of the scan, and records what happened afterwards so the next recommendation is better.

Inputs

1. VIPER (or any inspection) payload: VIN, 360 exterior images, undercarriage images, damage list (part, type, severity, image reference), tire tread, timestamps. When VIPER is absent, the same schema is filled from Underscope's own photo flow (exterior, interior, engine bay, underbody).
2. Non-visual facts the scan cannot give: odometer, title brand and state, keys, runs-and-drives, engine start clip (ACV records 30 seconds of engine audio through startup, idle, two revs and shutoff for its reports), dash warning lights.
3. Seller profile: seller type, location, contracts and lanes available to them, preference weights (price vs speed vs risk), repair capability.
4. Market data feeds: clean values, salvage comps, parts prices, fee schedules, transport rates.

Outputs

1. Ranked list of available channels with expected net (P10, P50, P90), days to cash, completion probability and dispute risk.
2. Buyer segment most likely to pay in the winning channel, with the reason in one sentence.
3. Confidence: clear winner, close call, or insufficient information, plus the single check that would most change the ranking.

4. Evidence pack: the images, measurements and comparables behind the numbers, so a human or a buyer can audit it.
5. A decision record: inputs, model versions, recommendation, who approved it, what the seller chose, and the realized outcome when known.

Functional requirements

- Eligibility filter runs first: title brand, state rules, seller contract, export restrictions; excluded channels are shown as excluded with the reason.
- Every channel estimate is a distribution from comparables plus model adjustment, never a single predicted price.
- Repair-cost estimates carry an explicit band and a list of invisible-risk items (frame, mechanical, flood) that widen it.
- Human review is routed by rule: close calls, high-value vehicles, low-confidence vision output, and a random blind sample, 3% at the start and raised as bands calibrate, reviewed without seeing the model's answer, to measure what the model misses.
- Outcome capture is mandatory: the system cannot close a decision record without a realized channel and price or an explicit "unknown".
- Explanations are written for a used-car manager, not a data scientist: three reasons, two risks, one next step.
- Seller preferences change the ranking in real time and are stored with the decision record.

Non-functional requirements

- Latency: ranking in under 60 seconds after the scan; market data cached and refreshed daily.
- Scale: 1,000 vehicles a month in the pilot, designed for 100,000 a month (Copart's U.S. volume is several hundred thousand a month).
- Auditability: every number traceable to its inputs and model version for 7 years, matching the retention sellers' finance teams expect.
- Data boundaries: seller data never trains a model used for another seller's competitor without contract permission; buyer identities stay anonymized outside the platform owner.
- Portability: the engine runs as a service behind an API; Copart/ACV can host it inside their environment with their data never leaving.
- Model governance: a model cannot be promoted without passing the backtest harness in section 11 on held-out vehicles.

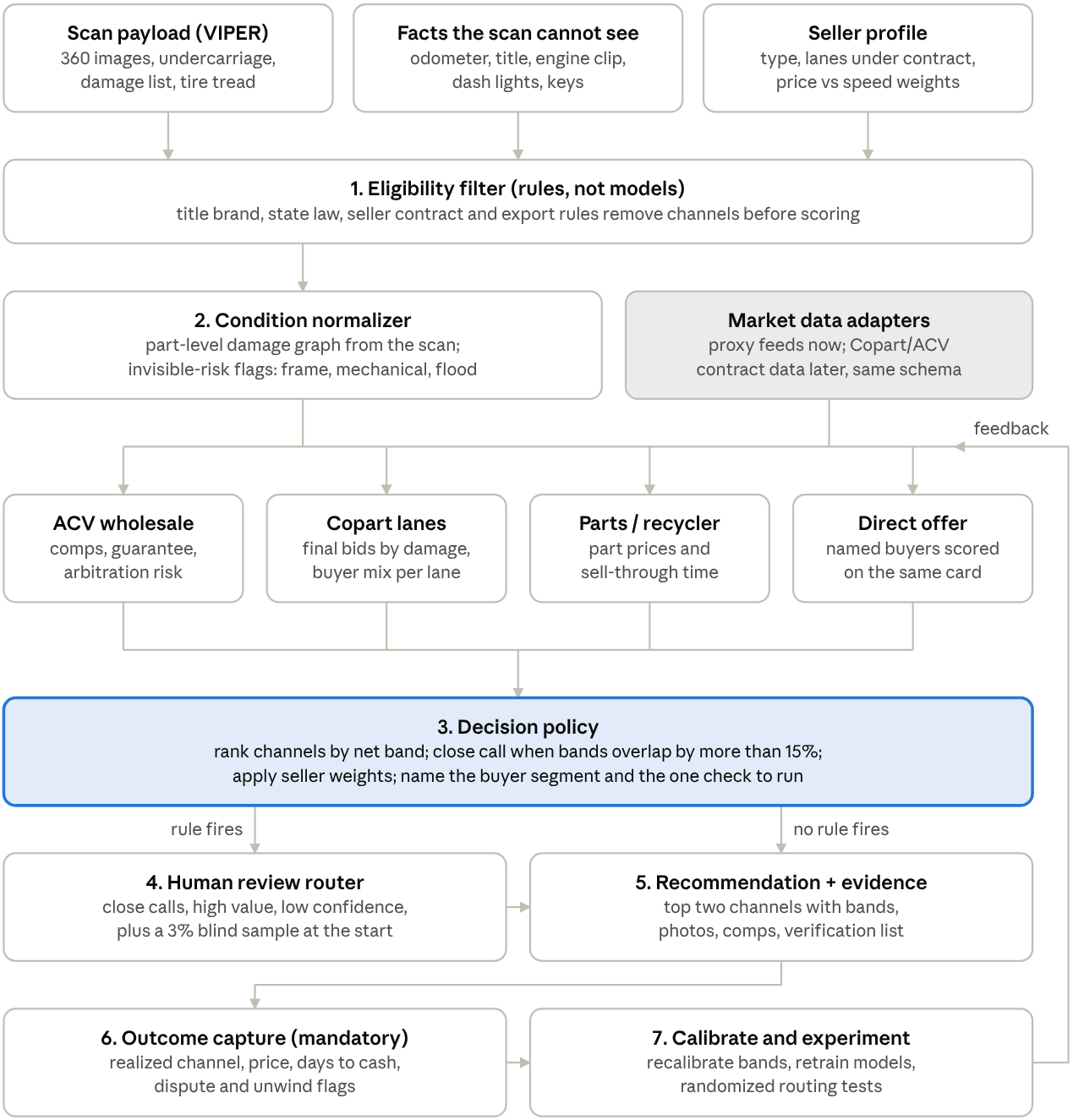
Out of scope for the first version: retail pricing for consumers, repair shop management, bidding automation on behalf of buyers, and any purchase of inventory by the platform.

Underlying logic and architecture

Rules decide what is allowed, estimators compute what each channel would pay, a policy ranks the channels under uncertainty, and realized outcomes feed back into the estimators. No stage is a single black-box prediction, and every stage can be replaced when Copart/ACV data arrives without changing the stages around it.

Stage	Rules or model	Why
Eligibility filter	Rules	Title law and contracts are facts, and a wrong channel here is a legal problem, not a statistical one
Condition normalizer	Model (vision) plus a fixed schema	Turns any scan into one part-level damage graph so estimators never see raw photos
Channel estimators	Comparables plus a small model per channel	Each channel has its own data and its own failure modes; separate models keep errors from leaking across channels
Decision policy	Deterministic over sampled outcomes	Monte Carlo over the estimators' bands gives the probability that each channel is best, which is what a close-call test needs
Human review router	Rules	Review budget is spent where it changes decisions or measures the model, never everywhere
Feedback	Calibration plus randomized experiments	Only a randomized slice reveals routing lift; observational data cannot

Rules filter, estimators price channels, a policy ranks, outcomes feed back



channel-routing engine · 7 stages, 4 estimators, 1 feedback loop

The scan and the seller's facts pass a rules-only filter, become a part-level condition graph, are priced by four independent estimators fed by swappable data adapters, ranked by one policy, reviewed by rule, delivered with evidence, and closed with a mandatory outcome that recalibrates the estimators.

How the policy ranks:

1. Draw 2,000 samples from each channel's net distribution (price band, days, dispute and unwind probabilities).
2. Compute the share of samples in which each channel wins; the recommended channel is the one with the highest share, not the highest mean.
3. If the top two shares are within the close-call threshold, compute which single input (frame check, engine clip, title confirmation) would most reduce the overlap; that becomes the verification item.
4. Apply seller weights (price, speed, risk) as a utility over the samples, then re-rank.
5. Name the buyer segment with the highest expected contribution inside the winning channel, with one reason.

Interfaces, so the engine can live inside Copart/ACV later:

- `POST /route` takes the scan payload, facts and seller profile; returns the ranking, bands, buyer segment, confidence and verification list.
- Market adapters are plug-ins behind one schema (`comps`, `values`, `parts`, `fees`, `transport`); the prototype ships proxy adapters, the pilot swaps in theirs.
- `POST /outcome` closes a decision record; a record cannot close without a realized channel and price or an explicit unknown.
- Every response carries model versions and an evidence pack id for audit.

Building without real data

The prototype proves the mechanism, not the prices: it runs end to end on proxy data, publishes bands whose coverage we can measure on the salvage side, and ships with a harness that shows exactly what changes when Copart/ACV data replaces the proxies. The sentence to Copart is "here is the engine, the schema and the test that measures lift; plug in your data and run it in shadow mode".

Contract field	Proxy in the prototype	Assumption to state out loud
Copart final price by lane, damage, title	Archived sold lots from an aggregator (tens of thousands of lots), plus members' CSV sales data	Archives are complete and unbiased for the models we pick; verify on a hand sample

Contract field	Proxy in the prototype	Assumption to state out loud
ACV sale price and bids	Clean wholesale values from a licensed API, adjusted by condition grade; a panel of 3 to 5 partner dealers sharing their own ACV sales history (sellers own their records)	Condition adjustments transfer across models; the panel is small and regional
Buyer segment of the winner	Lane buyer mix from Copart's own descriptions (dismantlers largest by volume, international buyers about 30% of units)	Mix is stable month to month
Arbitration and unwind	ACV policy terms (10-day window, 25% rate threshold) as bounds; Copart as-is	Rates per segment are unknown until the pilot
Days to cash	Lane calendars and auction formats	Payment timing by segment is unknown
Transport	Distance to nearest yard times a published per-mile rate	Rates vary by lane and season
Repair cost	Parts prices plus labor-hour heuristics, always as a band	Labor times are approximate until a shop partner supplies invoices

Simulation, used for the policy and never as a result:

- Fit a distribution per channel to the proxies, then simulate 10,000 vehicles with known "true" outcomes in every channel. This is the only place the counterfactual exists, so it is where the ranking policy, the close-call threshold and the review router are tuned.
- Run sensitivity on every parameter. The ones that flip the most decisions (expected: dispute cost in the guaranteed channel, rebuilt-title discount, days to cash for parts) become the first measurements of the pilot.
- Publish regret: how much net the policy gives up versus a perfect oracle, so the pilot has a number to beat.

The evaluation harness (a model is promoted only if it passes):

1. Condition: dollar-weighted recall of damage against blind human labels on our own scans; missing a $2,000 part counts twenty times a $100 scratch.
2. Price bands: on held-out Copart lots, the P10 to P90 band must cover 80% of realized prices and be no wider than a fixed-width baseline.
3. Ranking: on lots where the realized channel is known, the probability-of-winning estimate must be calibrated (when the engine says 70%, it is right about 70% of the time).
4. Plug-in: swapping an adapter for a partner dataset requires zero code changes and a re-run of tests 1 to 3.
5. Shadow mode at the pilot: the engine recommends silently next to the current lane rules; agreement rate, and lift measured only on a randomized 10% slice where the engine's choice is followed.

What we need from them to switch out of proxy mode is the contract in section 7; what we need from no one is everything above.

Operational costs

Running the engine costs about $2 to $5 per vehicle at pilot volume and under $1 per vehicle at scale, and more than 60% of the scaled cost is human review, not AI. The design lever is therefore the review router, and the breakeven bar is low: at 100,000 vehicles a month the engine pays for itself if it adds about $0.70 of net per vehicle scanned, for example $14 more on one vehicle in twenty.

Line	Basis	Per vehicle	1,000 a month (pilot)	100,000 a month (scale)
Vision on 30 photos, API model	Haiku 5.5 at $0.10 per million input tokens for prompts under 100K, about 1,568 tokens per photo (pricing, vision docs); Sonnet 5.5 at $2 input and $10 output is about $0.11	$0.006	$6	$600
Vision, self-hosted fine-tuned model	A published run served a fine-tuned 8B model on one A10G at about $0.003 per 98 images (Oxen); idle GPU time dominates at low volume	$0.001 plus about $600 a month per GPU	$600	$1,200 (two GPUs)
Explanation text	Short structured output, about 2,000 tokens on Haiku 5.5	$0.001	$1	$100

Line	Basis	Per vehicle	1,000 a month (pilot)	100,000 a month (scale)
Human review	20% of vehicles routed at 3 minutes plus a 3% blind sample at 5 minutes, about 0.75 minutes average, priced at the BLS median for auto damage appraisers, $78,240 a year or about $0.63 a minute (BLS via Wagedex)	$0.47	$470	$47,000 (about 8 full-time reviewers)
Market data licenses	Value, parts and archive feeds; unverified estimate pending vendor quotes	varies	$1,000 to $3,000	$5,000 to $20,000
Infrastructure	API, storage, database, monitoring; unverified estimate	varies	$500	$10,000
Total (API vision)		about $2 to $5	$2,000 to $4,000	$63,000 to $78,000, about $0.63 to $0.78 per vehicle

Notes on the numbers:

- Engineering and data-science salaries are excluded; they are the real cost of the first year and do not scale with vehicles.
- The review mix is a design choice. The blind sample starts at 3%; raising it to 7% would add about $13,000 a month at scale but sharpen the measurement of what the model misses; raise it if the harness shows calibration drifting.

- If Copart/ACV host the engine, their existing inspection and arbitration staff can absorb the review minutes, which turns the largest line into a reassignment rather than new spend.
- Per-vehicle AI cost is small enough that model choice should follow accuracy on the harness, not price.

Production readiness, risks and open questions

Verdict: not ready for production as a standalone product, and it should not try to be; it is ready for a shadow-mode pilot inside one partner's workflow in about 90 days if scope narrows to one seller segment (fleet, rental or finance units with damage) and one region. Copart/ACV are the natural host: their deal rationale names the capability, ACV stays a separate subsidiary with its own site, and the integration they have announced covers buyer liquidity, logistics and selected technology, not a cross-channel decision layer.

Whether they would bring it to production depends on three things we cannot know from outside: who owns the "combined demand engine" inside Copart, whether they prefer to buy capability (Monk in 2022, ACV in 2026) or license it, and how much integration bandwidth remains before fiscal 2028, when they expect the deal to be accretive. The pilot is designed to answer all three.

Risk	Why it matters	Test that settles it
Copart builds it internally	The 10 Sep 2026 release promises the right vehicle to the right buyer; they bought Monk and ACV rather than partnering	Propose a 90-day shadow pilot; if declined, the engine is channel-agnostic and the fleet and rental segment is the fallback customer
Routing lift is small	Most vehicles obviously belong where they go; the engine only earns on contestable ones	Measure the contestable share in proxy data (vehicles whose channel bands overlap); if under 5% of volume with lift under $100 each, stop
Data terms	Commercial training on scraped auction archives may breach site terms	Lawyer's read of Copart and ACV terms; prefer partner-supplied data and sellers' own records

Risk	Why it matters	Test that settles it
Wrong recommendation causes a loss	Sellers will blame the tool; guarantees need a balance sheet	Decision-support terms, errors-and-omissions cover, shadow mode first, no price guarantee without a partner underwriting it
Title and state law	Non-repairable titles cannot be rebuilt in Texas or Nevada; rules differ by state	Encode rules for the ten highest-volume states before the pilot; a vehicle with an unknown rule is routed to review
Vision misses subtle damage	Dents and cracks score 27 to 40 and 9 to 17 AP on the public CarDD benchmark against 86 to 90 for glass and tires	Dollar-weighted recall gate in the harness; invisible-risk flags widen bands instead of pretending
No access to VIPER output	VIPER is ACV's hardware and payload	Build against our own capture using the same schema; the pilot maps VIPER fields to it
Objective conflict	Routing for seller net can move fees between Copart lanes and ACV	Make the objective an explicit parameter (seller net, platform revenue, blended) and report both
Operations creep	Vroom-style losses came from owning inventory and reconditioning	The platform never takes title to a vehicle

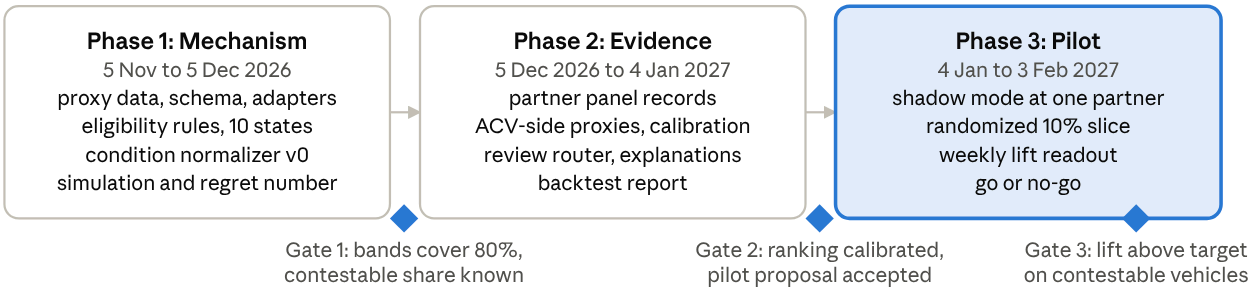
Open questions the next conversations must answer:

- [ ] Who inside Copart owns post-merger channel decisions, and is a third-party engine acceptable to them?
- [ ] What share of Blue Car and dealer consignments are contestable between lanes today, by Copart's own numbers?
- [ ] Will ACV expose the VIPER payload to a partner under NDA for a pilot?
- [ ] What arbitration and unwind rates by buyer segment can they share in aggregate?
- [ ] Which three partner dealers or one fleet will share their own sales records to seed the ACV-side proxies?

90-day plan and the questions for Copart/ACV

The goal of the 90 days is one sentence we can say to Copart with evidence: "we built the routing mechanism, it is calibrated on the salvage side, and here is the test that measures lift once your data is plugged in". The plan has three phases, each with a gate that can stop the project.

Three 30-day phases, each closed by a gate, end in a shadow-mode pilot



90-day plan · 3 phases, 3 gates

Phase 1 builds the mechanism on proxies, Phase 2 adds partner evidence and calibration, and Phase 3 runs it silently beside a partner's current rules; each gate must pass before the next phase starts.

Pilot design to propose at Gate 2:

- Metric: realized net proceeds per contestable vehicle, engine-routed versus rule-routed, with days to cash and dispute rate as secondary measures.
- Sample: about 1,600 vehicles per arm if the expected lift is $150 against a $1,500 spread in net (80% power, 5% significance); recompute once the real spread is known.
- Duration: 30 days in shadow mode, then 30 days with a randomized 10% slice following the engine.
- Scope: one lane (fleet, rental or finance units with damage), one region, one objective (seller net), reported weekly.

Questions to put to Copart/ACV, in the order to ask them:

1. Who owns the "right vehicle to the right buyer through the right channel" capability after the merger, and what have they already decided to build?
2. What share of Blue Car and dealer consignments do they see as contestable between salvage, wholesale and ACV lanes?

3. Would they run a 60-day shadow-mode pilot on one lane under NDA, with the data contract in section 7 as the input?
4. Can ACV share the VIPER payload schema and a sample of anonymized scans?
5. What aggregate arbitration, unwind and days-to-cash figures by buyer segment can they share to replace our proxies?
6. Would they prefer to license the engine, host it, or acquire it, and what evidence would move them from pilot to production?

What we want to achieve, stated so it can fail: a calibrated engine, a measured contestable share, a partner pilot under way, and a decision by 3 Feb 2027 on whether the lift justifies production. If Copart/ACV decline, the same engine and evidence go to fleet and rental remarketers, whose volume into Copart grew about 20% in the latest quarter.

Sources

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